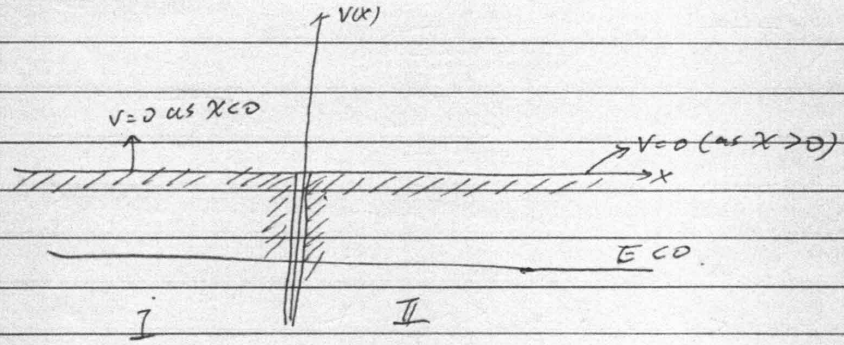
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(莊道茂)

Ex 1. First, to find sol. of $V = -g\delta(x)$ and then $\xrightarrow{x'=x-x_0}$ $V = -g(\delta(x-x_0))$

① If $E \gg V$ ⇒ like free particle. → plane wave.

② If $E < V$



由 Time independent Schrödinger's equation.

$$\frac{-\hbar^2}{2m}\frac{\partial^2}{\partial x^2} + V(x)\psi(x) = E\psi(x) \quad \text{——①}$$

I: $k_1 = \frac{\sqrt{2m(V_0-E)}}{\hbar} = \frac{\sqrt{-2mE}}{\hbar}$, $E<0$

$$\psi_1 = Ae^{kx}, \quad x<0, \quad \text{——②}$$

II: $k_2 = k_1 \Rightarrow \psi_2 = Be^{-kx}$, $x>0$

$\because \psi_1(0) = \psi_2(0) \therefore A = B$. ——③

由① $\psi_2'(0) - \psi_1'(0) = -\frac{2mg}{\hbar^2}\psi(0) = -\frac{2mg}{\hbar^2}A$

$$\Rightarrow \left(-kAe^{-kx} - kAe^{kx}\right)\Big|_{x=0} = \frac{-2mg}{\hbar^2}A$$

$$\Rightarrow -2kA = \frac{-2mg}{\hbar^2}A$$

$$\Rightarrow k = \frac{mg}{\hbar^2} = \frac{\sqrt{-2mE}}{\hbar^2}$$

$$\Rightarrow E = -\frac{mg^2}{2\hbar^2}$$ (唯一解, δ potential only one bound state) timeless

Normalized condition :

$$1 = \int_{-\infty}^{\infty} \psi^*(x)\,\psi(x)\,dx$$

$$= \int_{-\infty}^{0} \psi_1^*(x)\,\psi_1(x)\,dx + \int_{0}^{\infty} \psi_2^*\psi_2\,dx$$

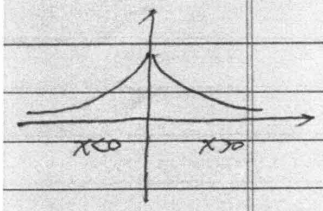
$$= 2\int_{0}^{\infty} \psi_2^*\psi_2\,dx = 2\int_{0}^{\infty} A^2 e^{-2kx}\,dx$$

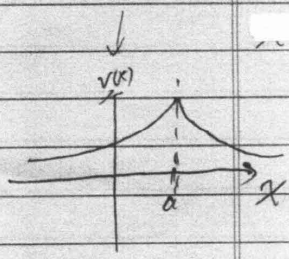
$$= 2A^2\left(\frac{-1}{2k}\right) e^{-2kx}\Big|_0^{\infty}$$

$$-\frac{A^2}{k}$$

$$\Rightarrow A = \sqrt{k} = \frac{\sqrt{mg}}{\hbar}$$

$$\Rightarrow \psi(x) = \begin{cases} \frac{\sqrt{mg}}{\hbar} e^{-(mg/\hbar^2)x}, & x > 0 \\ \frac{\sqrt{mg}}{\hbar^2} e^{(mg/\hbar^2)x}, & x < 0 \end{cases}$$





Absolute value.

$$\Rightarrow \boxed{\psi(x') = \frac{\sqrt{mg}}{\hbar} e^{-\left(\frac{mg}{\hbar^2}\right)|x'|}}$$

$$\Rightarrow \boxed{\psi(x) = \frac{\sqrt{mg}}{\hbar} e^{-\left(\frac{mg}{\hbar^2}\right)|x-a|}}$$ — g- representa

timeless

<P-representation>

$$\psi(p) = \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{\infty} dx' \left(\sqrt{\frac{mg}{\hbar}}\right) e^{-\left(\frac{mg}{\hbar^2}\right)|x'-a|} e^{-\frac{ip}{\hbar}x'}$$

$$= \left(\frac{1}{\sqrt{2\pi\hbar}}\sqrt{\frac{mg}{\hbar}}\right)\left\{\int_{-\infty}^{0} dx' e^{\left(\frac{mg}{\hbar^2} - \frac{ip}{\hbar}\right)x' - \frac{mga}{\hbar^2}} + \int_{0}^{\infty} dx' e^{\left(-\frac{mg}{\hbar^2} - \frac{ip}{\hbar}\right)x' + \frac{mga}{\hbar^2}}\right\}$$

$$\left(\text{let } C = \frac{-mga}{\hbar^2},\ D = \frac{mga}{\hbar^2},\ E = \frac{mg}{\hbar^2} - \frac{ip}{\hbar},\right.$$
$$\left. F = \frac{mg}{\hbar^2} + \frac{ip}{\hbar},\ G = \frac{1}{\sqrt{2\pi\hbar}}\sqrt{\frac{mg}{\hbar^2}}\right)$$

$$\Rightarrow \psi(p) = G\left\{C\int_{-\infty}^{0} dx' e^{Ex'} + D\int_{0}^{\infty} dx' e^{-Fx'}\right\}$$

$$= G\left\{\frac{C}{E} e^{Ex'}\Big|_{-\infty}^{0} + \left(\frac{-D}{F}\right) e^{-Fx'}\Big|_{0}^{\infty}\right\}$$

$$= G\left\{\left(\frac{C}{E}\right) + \left(\frac{D}{F}\right)\right\}$$

$$= \left(\frac{1}{\hbar}\sqrt{\frac{mg}{2\pi\hbar}}\right)\left\{\frac{-mga/\hbar^2}{\frac{mg}{\hbar^2} - \frac{ip\hbar}{\hbar^2}} + \frac{mga/\hbar^2}{\frac{mg}{\hbar^2} + \frac{ip\hbar}{\hbar^2}}\right\}$$

$$= \frac{1}{\hbar}\sqrt{\frac{mg}{2\pi\hbar}}\left\{\frac{mga}{mg + ip\hbar} - \frac{mga}{mg - ip\hbar}\right\}$$

$$= \frac{mga}{\hbar}\sqrt{\frac{mg}{2\pi\hbar}}\left[\frac{-2ip\hbar}{(mg)^2 + (p\hbar)^2}\right]$$

$$= (mga)\sqrt{\frac{mg}{2\pi\hbar}}\left(\frac{-2ip}{(mg)^2 + (p\hbar)^2}\right) \text{ or } = \sqrt{\frac{mg}{2\pi\hbar}}\left(\frac{-2ip\,mga}{(mg)^2 + (p\hbar)^2}\right)$$

timeless

step { 1. derive part of proof process for uncertainty principle
2. derive minimum state.

uncertainty ⇒ Hermitian

Ex. 2. /10 1. For any observable $\hat{A}$, $\hat{A} = \hat{B}\hat{B}^\dagger$. The mean value in a state ψ of a physical quantity

$$\Rightarrow \langle \hat{A} \rangle = \langle BB^\dagger \rangle = \langle B^\dagger \psi | B^\dagger \psi \rangle \geq 0 \quad \text{— (1)}$$

set $\hat{C}, \hat{D}$ are two another Hermitian operators and $\lambda \in \mathbb{R}$.

Then, let $\hat{B} = \hat{C} + i\lambda\hat{D}$, $\hat{B}^\dagger = \hat{C} - i\lambda\hat{D}$

$$\Rightarrow \langle \hat{A} \rangle = \langle \hat{C}^2 \rangle + \lambda^2 \langle \hat{D}^2 \rangle - i\lambda \langle \hat{C}\hat{D} - \hat{D}\hat{C} \rangle$$
$$\equiv f(\lambda)$$

Obtain f_{min}, by $\frac{df}{d\lambda} = 0$.

$$\Rightarrow f_{min} = \langle \hat{C}^2 \rangle + \frac{\langle \hat{C}\hat{D} - \hat{D}\hat{C} \rangle^2}{4\langle \hat{D}^2 \rangle} \geq 0 \quad \text{— (2)}$$

Therefore,

$$\langle \hat{C}^2 \rangle \langle \hat{D}^2 \rangle \geq \frac{1}{4} \langle \hat{C}\hat{D} - \hat{D}\hat{C} \rangle^2$$

Let $\hat{C} = \delta\hat{M} = \hat{M} - \langle M \rangle$, $\hat{D} = \delta\hat{N} = \hat{N} - \langle N \rangle$ (deviation) and $[\delta\hat{M}, \delta\hat{N}] = [\hat{M}, \hat{N}]$

$$\Rightarrow \langle (\delta\hat{M})^2 \rangle \langle (\delta\hat{N})^2 \rangle \geq \frac{1}{4} \langle \hat{M}\hat{N} - \hat{N}\hat{M} \rangle^2$$

let $\Delta M = \left((\delta\hat{M})^2 \right)^{1/2}$, $\Delta N = \left((\delta\hat{N})^2 \right)^{1/2}$

$$\Rightarrow \boxed{\Delta M \Delta N \geq \sqrt{\frac{1}{4} \langle [\hat{M}, \hat{N}] \rangle^2}} \quad \text{— (3)}$$

★★★ 2. The equality hold only if

$$\hat{B}^\dagger \psi(x) = (\hat{C} - i\lambda\hat{D})\psi(x) = 0 \quad \text{— (4)}$$

timeless

Similar discussion in

* "Griffiths' QM book, to see P.113, [3.66]"

Then $\lambda = \frac{i}{2}\frac{\langle[\hat{C},\hat{D}]\rangle}{\langle\hat{D}^2\rangle}$ as $f_{min.}(\lambda) = f(\lambda')$ (5)

(5) Substitute into (4)

$$\Rightarrow \left(\hat{C} + \frac{1}{2}\frac{\langle[\hat{C},\hat{D}]\rangle}{\langle\hat{D}^2\rangle}\hat{D}\right)\psi(x) = 0$$

$\hat{C} \to \hat{M}$, $\hat{D} \to \hat{N}$

$$\Rightarrow \left[(\hat{M} - \langle M\rangle) + \frac{1}{2}\frac{\langle[\hat{M},\hat{N}]\rangle}{(\Delta N)^2}(\hat{N} - \langle N\rangle)\right]\psi(x) = 0$$

For special case, $\hat{M} = \hat{p}$, $\hat{N} = \hat{x}$.

$$\Rightarrow \left[\left(-i\hbar\frac{d}{dx} - \langle p\rangle\right) + \frac{i\hbar}{2}\frac{(x - \langle x\rangle)}{(\Delta x)^2}\right]\psi(x) = 0$$

$$\Rightarrow \left(\frac{\hbar}{i}\frac{d}{dx} - \langle p\rangle\right)\psi = i\underbrace{\left(\frac{\hbar}{2(\Delta x)^2}\right)}_{a}(x - \langle x\rangle)\psi$$

a in "Griffiths" book!

$$\Rightarrow \boxed{\left(\frac{\hbar}{i}\frac{d}{dx} - \langle p\rangle\right)\psi = ia(x - \langle x\rangle)\psi}$$

⊙ Solving this equation:

$$\Rightarrow \frac{d\psi}{dx} = \frac{i}{\hbar}(iax - ia\langle x\rangle + \langle p\rangle)\psi$$

$$= \frac{a}{\hbar}\left(-x + \langle x\rangle + \frac{i}{a}\langle p\rangle\right)\psi$$

timeless

$$\Rightarrow \int \frac{d\psi}{\psi} = \frac{a}{\hbar}\int\left(-x+\langle x\rangle + \frac{i\langle p\rangle}{a}\right)dx$$

$$\Rightarrow \ln\psi = \frac{a}{\hbar}\left(-\frac{x^2}{2} + \langle x\rangle x + \frac{i\langle p\rangle}{a}x\right) + \text{Const.}$$

let Const. $= \underbrace{\frac{-\langle x\rangle^2 a}{2\hbar}}_{\text{"}} + \underbrace{S}_{\text{now const.}}$

$$\frac{-\langle x\rangle^2}{2\hbar}\cdot\frac{\hbar}{2(\Delta x)^2}$$

$$\Rightarrow \ln\psi = \frac{-a}{2\hbar}(x-\langle x\rangle)^2 + \frac{i\langle p\rangle}{\hbar}x + S$$

$a = \hbar/2(\Delta x)^2$

$$\Rightarrow \psi(x) = \left(e^{\frac{-a}{2\hbar}(x-\langle x\rangle)^2 + i\frac{\langle p\rangle}{\hbar}x}\right)\underbrace{N'}_{\text{"}}\underbrace{e^S}_{\text{"}}$$

$$= N' \exp\left[\frac{-(x-\langle x\rangle)^2}{4(\Delta x)^2} + i\left(\frac{\langle p\rangle}{\hbar}\right)x\right]$$

$\therefore$ Normalization condition

$$\int_{-\infty}^{\infty}|\psi(x)|^2\,dx = 1$$

$$\Rightarrow N' = \sqrt{\frac{1}{\sqrt{\hbar\pi}(\Delta x)^2}} \text{ or } = \sqrt{\frac{1}{\Delta x\sqrt{2\pi}}}$$

(Method I)

3. In momentum space, change $\hat{M} = \hat{x}$, $\hat{N} = \hat{p}$, $p - \langle p\rangle$

Then, we can simply derive (by symmetry)

$$\phi(p) = \sqrt{\frac{1}{\Delta p\sqrt{2\pi}}}\exp\left[-\frac{(p-\langle p\rangle)^2}{4(\Delta p)^2} - i\left(\frac{p}{\hbar}\right)x\right]$$

timeless